

Project Plan: Superior Open IP Intelligence Service
(Ready for handoff to Grok 4.6 for implementation)

1. Project Vision & Goals

Build a **high-quality, freely redistributable (or lightly attributed) IP intelligence database + service** that combines the best free/open sources into one superior product.

Core capabilities the final product must deliver:

- Country + City-level geolocation (with coordinates, timezone, subdivisions)
- ASN + organization name + domain
- Datacenter / Hosting / Cloud detection
- VPN / Proxy / Tor / Relay / CDN / Anonymous detection
- Clean MMDB output (primary) + CSV/Parquet
- Daily automated updates
- Simple lookup API + offline library support
- Target final MMDB size: **80–150 MB** (sweet spot)

Success criteria:

- Better combined coverage than any single free source
- Drop-in compatible with standard MaxMind MMDB readers
- Fully automated daily rebuild + release
- Clear attribution & license compliance
- Easy for others to self-host or consume

2. Data Sources Inventory (Priority Order)

Priority	Source	What it provides	Format	Update	License / Notes	Role in merge
1	IPinfo Lite	Country + Continent + ASN + AS name + domain	MMDB/CSV	Daily	CC BY-SA 4.0 (attribution required)	Primary ASN country fallback

2	GeoLite2-City	Full city geo (names multi-lang, lat/lon, timezone, subdivisions)	MMDB	2×/week	CC BY-SA 4.0 + MaxMind EULA (attribution + account needed)	Primary geo data
3	GeoLite2-ASN	ASN fallback	MMDB	2×/week	Same as above	ASN fallback
4	DB-IP City Lite	Supplementary city data	MMDB	Monthly	CC BY 4.0	Geo enrichment / conflict resolution
5	sapics/ip-location-db (user-country , origin-asn)	Clean country + ASN from public data	CSV/MMDB	Daily	PDDL (public domain – best)	High-priority country/ASN when available
6	iptoasn.com	IP→ASN + country	TSV	Hourly	Free	ASN enrichment
7	OpenProxyDB (NetworkCats)	Proxy / VPN / Tor / hosting / CDN / school flags	CSV	Daily	CC0	Core privacy/hosting flags
8	ipapi.is free geo	IPv4/IPv6 geolocation	CSV	Frequent	Free download available	Additional geo
9	Cloud provider ranges (AWS, GCP, Azure, DigitalOcean, Hetzner, OVH, Cloudflare, etc.)	Official hosting ranges	JSON/CSV	Daily	Public	Strong is_hosting / is_cdn signals

10	Public VPN lists (NordVPN servers, X4B, iCloud Private Relay, Tor exits, etc.)	Known VPN/relay exits	Various	Daily	Mixed (mostly public)	is_vpn / is_relay overlay
11	QQWry / Chunzhen (optional)	Better China coverage	Binary/CSV	—	Check license	China-specific enrichment

Note: Prefer sources that already publish MMDB or clean CIDR CSVs. Avoid single-IP residential proxy dumps that explode size.

3. Target Output Schema (MMDB)

Keep it close to MaxMind + IPinfo style for compatibility:

JSON

```

{
  "country": { "iso_code": "US", "names": { "en": "United States" } },
  "continent": { "code": "NA", "names": { "en": "North America" } },
  "city": { "names": { "en": "..." }, "geoname_id": ... },
  "location": {
    "latitude": ...,
    "longitude": ...,
    "accuracy_radius": ...,
    "time_zone": "..."
  },
  "subdivisions": [...],
  "postal": { "code": "..." },
  "asn": {
    "autonomous_system_number": 15169,
    "autonomous_system_organization": "Google LLC",
    "as_domain": "google.com"
  },
  "traits": {
    "is_anonymous": false,
    "is_anonymous_vpn": false,
    "is_hosting_provider": true,
    "is_public_proxy": false,
    "is_tor_exit_node": false,
    "is_cdn": false,
    "is_relay": false
  }
}

```

Add only fields that are reliable. Drop multi-language names if size becomes an issue (keep English only).

4. Architecture & Tech Stack

Recommended stack (simple + robust for Grok 4.6):

- **Language:** Go (best ecosystem for MMDB – use github.com/maxmind/mmdbwriter + github.com/oschwald/maxminddb-golang)
- Alternative: Python (faster to prototype) + maxminddb + custom writer, or use existing NetworkCats tool as base
- **Merge engine:** Extend or fork **NetworkCats/Merged-IP-Data** (already does priority field-level merging + OpenProxyDB integration)
- Or use official **mmdbconvert** (MaxMind) for clean non-overlapping merges

- **Pipeline:** GitHub Actions (daily) or simple cron + Docker
- **Storage:** GitHub Releases for the MMDB + optional S3/R2 mirror
- **Serving (Phase 2):**
 - Lightweight Go/Rust API (`rustyip` style or simple Gin/FastAPI)
 - Local library wrappers
 - Optional Redis/memory-mapped cache for high QPS

Merge Strategy (critical):

1. Download all sources.
2. Normalize everything to CIDR + structured records.
3. Resolve overlapping networks → most specific prefix wins.
4. Field-level priority (configurable):
 - Geo: GeoLite2-City > DB-IP > others
 - ASN: IPinfo Lite > sapics/iptoasn > GeoLite2-ASN
 - Flags: OR logic for booleans (`is_vpn` = true if *any* source says so)
5. Collapse adjacent identical records.
6. Write single optimized MMDB (record size 24 or 28).
7. Generate checksums + attribution file.

Size control tactics:

- Prefer CIDR ranges over individual IPs
 - English-only names
 - Drop low-value fields early
 - Monitor size after each merge; abort if > 180 MB
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5. Implementation Phases (for Grok 4.6)

Phase 0 – Setup (1–2 days)

- Create repo
- Decide base: fork NetworkCats/Merged-IP-Data or start fresh with mmdbwriter
- Set up MaxMind account + license key (for GeoLite2)
- Document all licenses & required attributions

Phase 1 – MVP Data Pipeline (core)

- Automated downloaders for every source
- Basic merge → single MMDb with country + ASN + basic flags
- GitHub Action that runs daily and publishes to Releases
- Validation script (sample IPs: 8.8.8.8, 1.1.1.1, known VPN/hosting IPs)
- Target size < 100 MB

Phase 2 – Full Enrichment

- Add city-level data + coordinates
- Full OpenProxyDB + cloud ranges + VPN lists
- China enhancement (optional)
- Size optimization pass
- Attribution README auto-generated

Phase 3 – Service Layer

- Simple HTTP API (lookup by IP → JSON)
- Rate limiting + caching
- Docker image + docker-compose
- Client libraries (Go, Python, JS) or just document MMDb usage
- Optional: Prometheus metrics, health endpoint

Phase 4 – Quality & Operations

- Automated accuracy checks against known ground truth
 - Diff reports between daily builds
 - Size & coverage dashboards
 - Mirror strategy (jsDelivr / Cloudflare R2 / GitHub)
 - Clear “How to use / License / Attribution” page
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6. Licensing & Compliance Rules (Non-negotiable)

- Always ship a clear `ATTRIBUTION.md` and `LICENSE` that lists every source
- GeoLite2 & IPinfo Lite require visible attribution
- Prefer PDDL / CC0 sources where possible
- Do **not** claim “commercial-grade accuracy”

- If redistributing commercially, review MaxMind EULA carefully
 - Never remove or hide source credits
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7. Success Metrics

- Daily successful build rate > 99%
 - Final MMDB size 80–150 MB
 - Lookup latency < 1 ms (local MMDB)
 - Coverage of major cloud providers + top VPN providers
 - Clear superiority over single free sources on combined signals
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8. Risks & Mitigations

Risk	Mitigation
Source license changes	Monitor + have fallback sources
Size explosion	Strict CIDR preference + field pruning
Conflicting data	Explicit priority config + logging of conflicts
MaxMind account/key issues	Cache last good GeoLite2 + use alternatives
False positives on VPN flags	OR logic + confidence later if needed
Maintenance burden	Fully automated pipeline + clear docs

9. Deliverables Checklist (for Grok 4.6)

- ☐ Working daily pipeline that produces `Superior-IP.mmdb`
 - ☐ GitHub Releases with MMDB + checksum + attribution
 - ☐ Configurable merge priority file
 - ☐ Basic lookup CLI + example code
 - ☐ Simple API server (optional but recommended)
 - ☐ Full documentation (sources, licenses, how to rebuild, how to consume)
 - ☐ Size & quality reports
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Recommended Starting Point for Grok 4.6

1. Fork or heavily reference <https://github.com/NetworkCats/Merged-IP-Data>
2. Add missing sources (sapics PDDL datasets, official cloud ranges, more VPN lists)
3. Improve priority configuration and size control
4. Add strong attribution generation
5. Publish as a clean new project (e.g. `superior-ip-db` or `open-ip-intel`)

This plan is deliberately practical, license-aware, and sized for real-world use. It produces something meaningfully better than any individual free database while remaining maintainable and legally usable.

Hand this entire document to Grok 4.6 and instruct it to begin with Phase 0 + Phase 1.